



LAB REPORT



COURSE INFORMATION

- ❖ Course Code:
 - ❖ Course Title:
 - ❖ Experiment No:
 - ❖ Experiment Name:
-



SUBMITTED TO

- ❖
 - ❖ Department of
 - ❖ East West University
-



SUBMITTED BY

- ❖ Student Name :
- ❖ ID :
- ❖ Section :
- ❖ Department :

East West University

Date of submission:

Data Collection:

(A) Length of the wire, $l =$ _____ cm

Calculation of the least count:

$$\text{Pitch (P)} = \frac{\text{The distance moved by plane of the disc along vertical scale}}{\text{Number of full rotations given to the circular scale}}$$

= _____

= _____

Number of total divisions in the circular scale, $n = 100$

$$\text{Least count (L.C.)} = \frac{\text{pitch}}{\text{Number of divisions of circular scale}} = \text{_____}$$

$$= \text{_____ mm}$$

$$= \text{_____ cm}$$

(B) Reading for the Radius of the Wire (r)

No. of obs.	Linear Scale Reading (x) cm	Circular Scale Divisions (CSD)	Least Count (LC) cm	Diameter $d = x + (\text{CSD} \times \text{LC})$ cm	Mean Diameter (d) cm	Mean Radius $r = d/2$ cm
01						
02						
03						

$$\text{Vernier Constant} = \frac{\text{One main scale division}}{\text{Number of divisions in vernier scale}} = \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}} \text{ mm}$$

$$= \underline{\hspace{2cm}} \text{ cm}$$

(C) Reading for the Radius of the Cylinder (a)

No. of obs.	Main Scale Reading (x) cm	Vernier Scale Division (VSD)	Vernier Constant (VC) cm	Diameter $D = x + (\text{VSD} \times \text{VC})$ cm	Mean Diameter D cm	Mean Radius $a = D/2$ cm
01						
02						
03						

(D) Reading for the Time Period (T)

No. of obs.	Time for 20 Oscillations (t) sec	Time Period $T = (t/20)$ sec	Mean Time Period T sec
01			
02			
03			
04			
05			

Calculation:

Mass of the cylinder, $M = \underline{\hspace{2cm}}$ gm

Moment of inertia of the cylinder, $I = \frac{1}{2} M a^2$

$$= \text{_____ gm-cm}^2$$

$$= \text{_____ gm-cm}^2$$

From data collection,

$$l = \text{_____ cm}, r = \text{_____ cm}, T = \text{_____ sec}, I = \text{_____ gm-cm}^2$$

So, we get from equation (3)

$$\eta = \frac{8\pi l l}{T^2 r^4}$$

$$= \text{_____ dynes/ sq.cm}$$

$$= \text{_____ dynes/ sq.cm}$$

Standard value (Theoretical value) of $\eta = 7.7 \times 10^{11}$ dynes/ sq.cm (Iron wire)

$$\text{Percentage Difference} = \frac{\text{Standard value} - \text{Experimental value}}{\text{Standard value}} \times 100\%$$

$$\text{Percentage difference} = \text{_____}$$

Result:

$$\text{The Modulus of rigidity of a wire is} = \text{_____ dynes/ sq.cm}$$